

Report: Organizational Units (OUs) & Structure

1. Introduction

A structured Organizational Unit layout was implemented in Active Directory to separate users and computers by department and function. This task set involved the creation of the Departments OU and its child OUs for Finance, HR, and IT, as well as a separate root-level OU for ServiceAccounts. User accounts previously located in the default Users container were relocated into their correct departmental OUs. A Group Policy Object was created and configured to restrict USB access for the Finance department. The GPO was linked to the Finance OU, and a domain-joined client machine was moved to that OU to confirm policy enforcement. Verification was performed using gpresult to ensure the policy was correctly applied to the target machine.

2. Objective

The objective was to apply structural organization to Active Directory by creating OUs that reflect departmental roles, moving user accounts to the correct OUs, and enforcing a department-specific policy. Tasks included creating the Departments OU and its sub-OUs, creating the ServiceAccounts OU, moving users into Finance, HR, and IT OUs, linking a USB-blocking GPO to the Finance OU, moving a client system into the same OU, and confirming policy application using system tools.

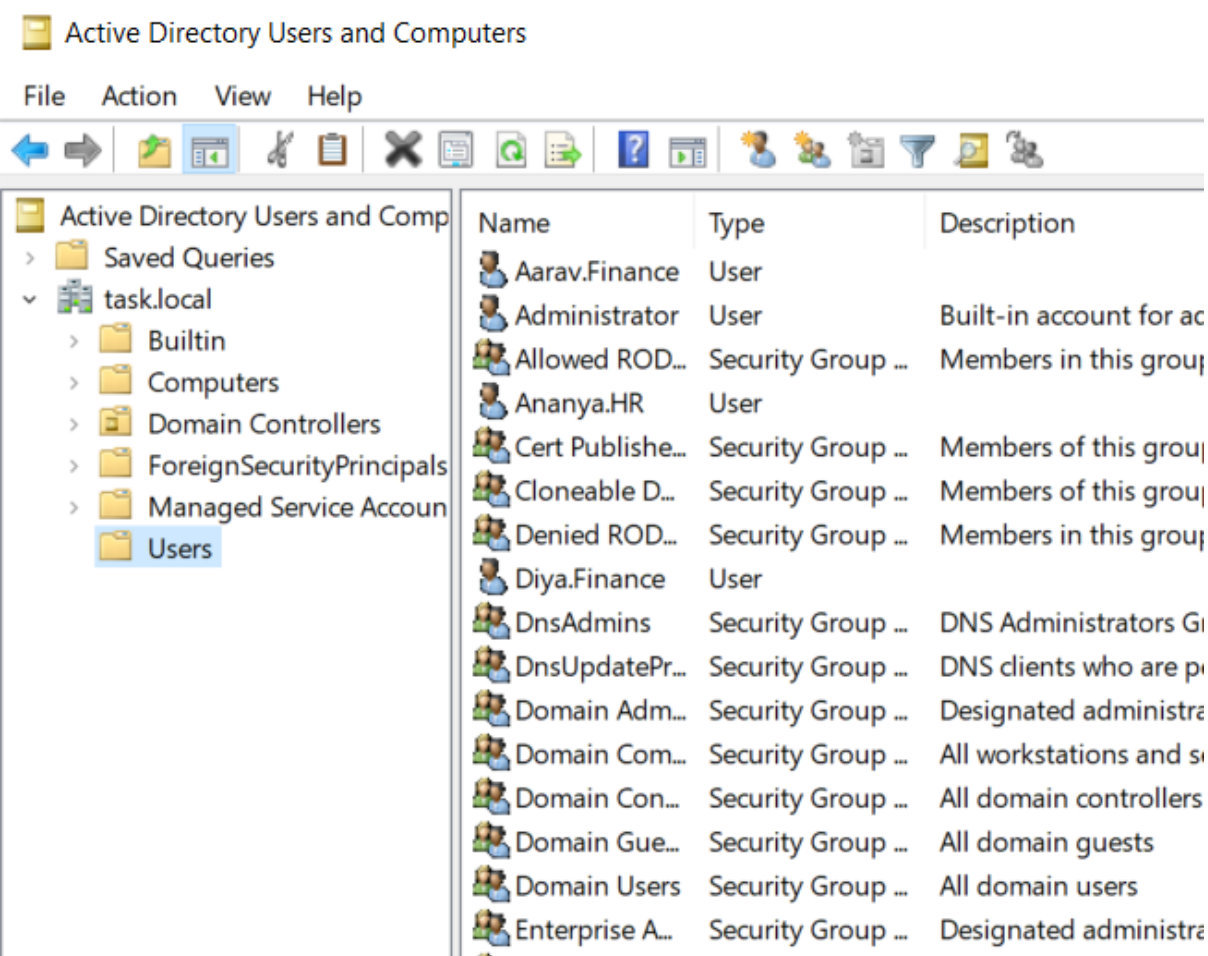
3. Tools used

- Windows Server (AD DS)
- Active Directory Users and Computers (ADUC)
- Group Policy Management Console (GPMC)
- Windows PowerShell
- Windows 10 Enterprise (domain joined)

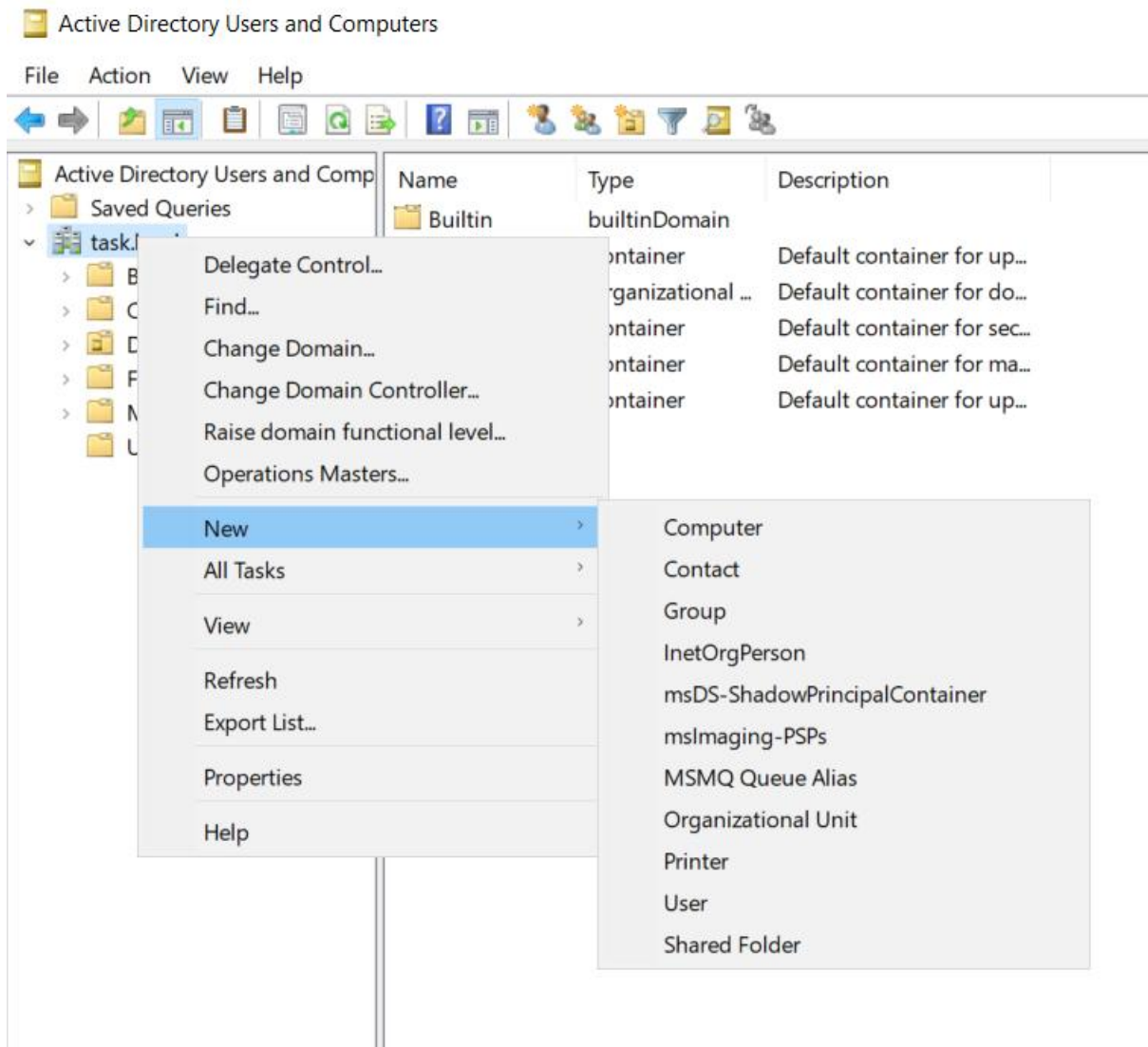
4. Methodology

4.1 Create Departmental Ous

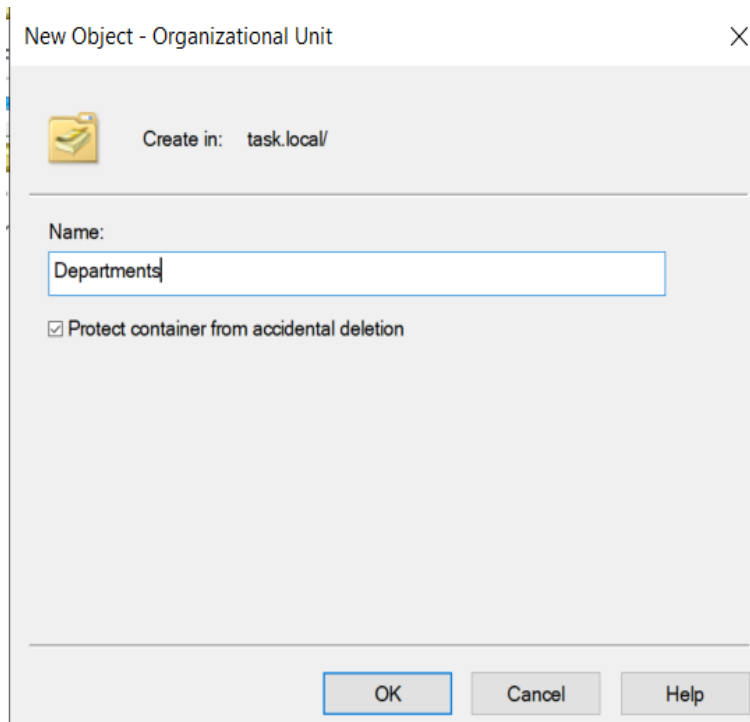
In ADUC, a top-level OU named Departments was created. Within it, three sub-OUs named Finance, HR, and IT were added. A separate OU named ServiceAccounts was created at the domain root level to hold system or service-related identities. Each OU was confirmed visible in the Active Directory structure.



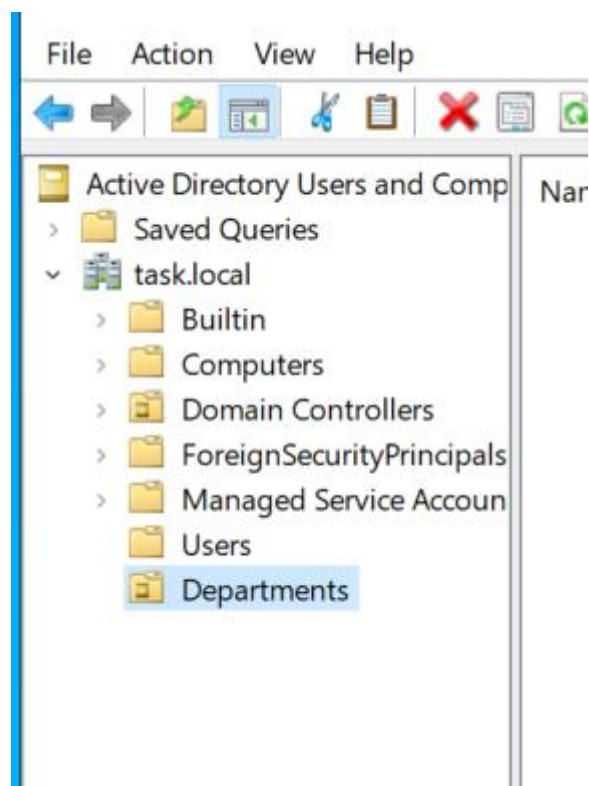
(No OU is there)



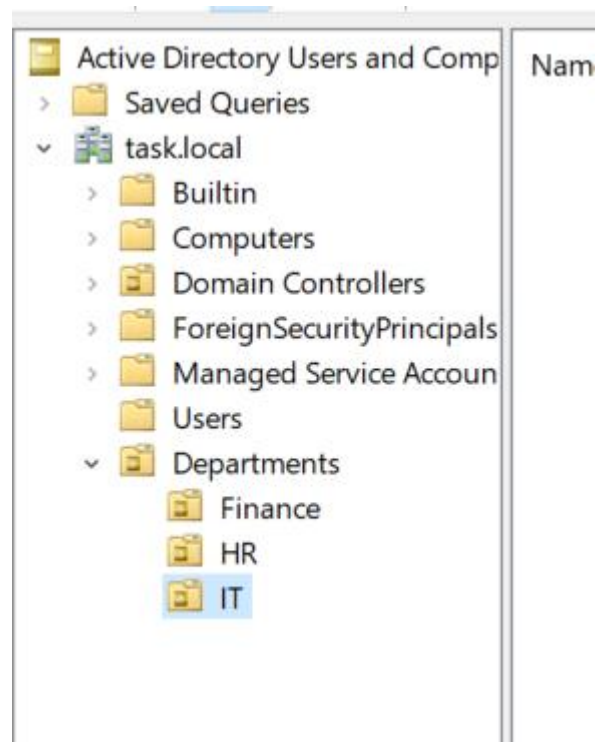
(Add new OU)



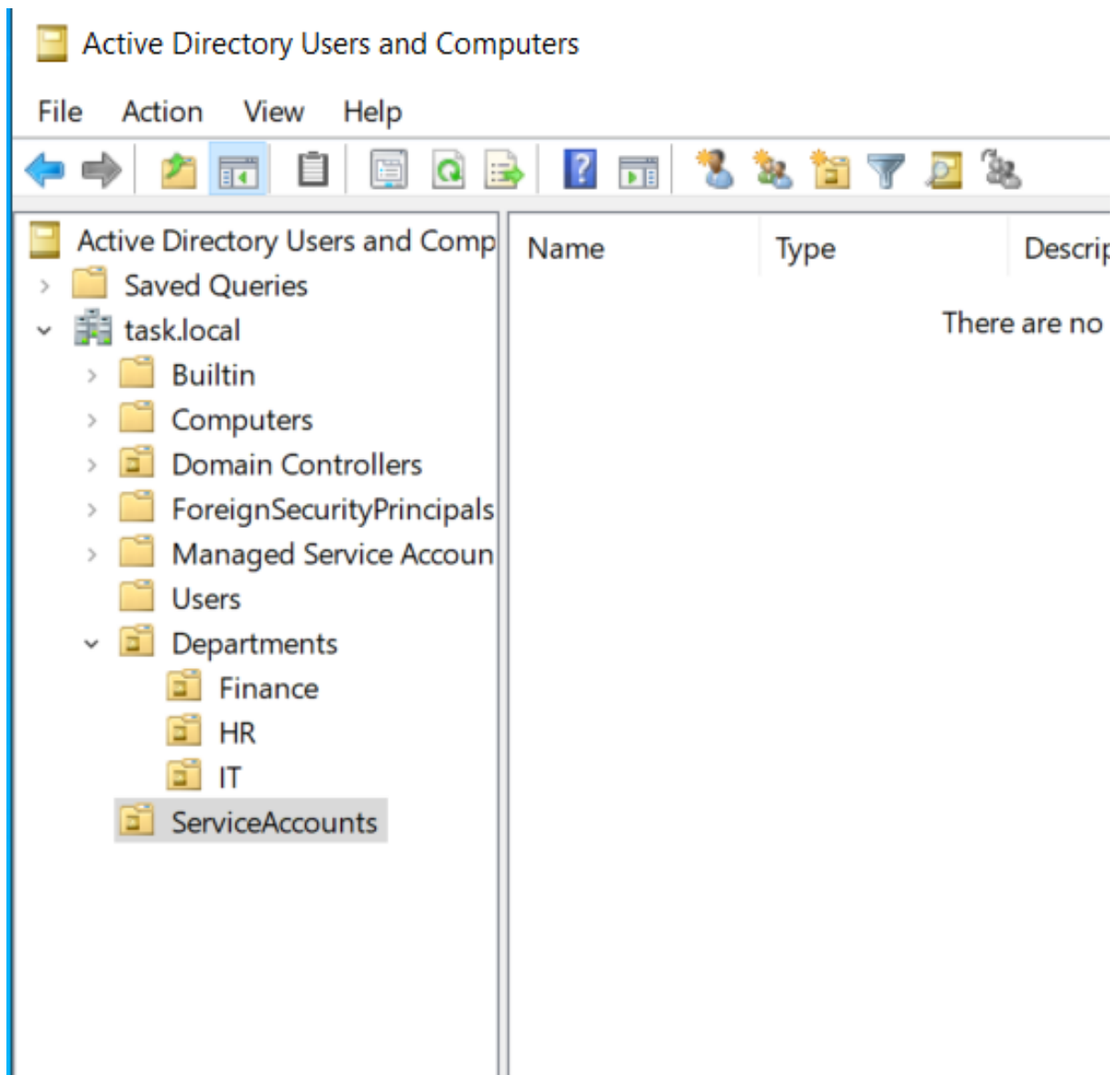
(Name it as departments)



(Successfully created a new OU)



(Added three child OU's)



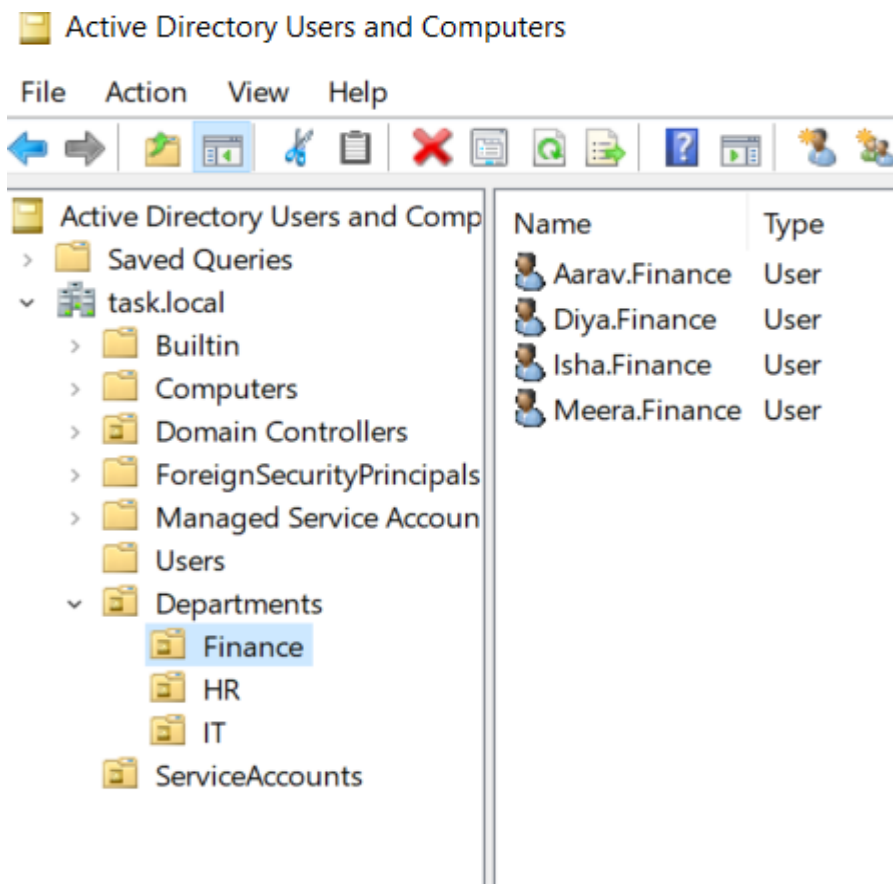
(Added a new OU – service accounts)

4.2 Move Users into Correct Departmental Ous

All user accounts that were initially present in the default Users container were relocated using PowerShell. The Move-ADObject command was used to move Finance users into the Finance OU, HR users into the HR OU, and IT users into the IT OU. After the command execution, each OU was opened in ADUC to verify the presence of the correct users.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> # Finance Users
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Aarav.Finance,CN=Users,DC=task,DC=local" -TargetPath "OU=Finance,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Diya.Finance,CN=Users,DC=task,DC=local" -TargetPath "OU=Finance,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Meera.Finance,CN=Users,DC=task,DC=local" -TargetPath "OU=Finance,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Isha.Finance,CN=Users,DC=task,DC=local" -TargetPath "OU=Finance,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator>
PS C:\Users\Administrator> # HR Users
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Rohan.HR,CN=Users,DC=task,DC=local" -TargetPath "OU=HR,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Ananya.HR,CN=Users,DC=task,DC=local" -TargetPath "OU=HR,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Sneha.HR,CN=Users,DC=task,DC=local" -TargetPath "OU=HR,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator>
PS C:\Users\Administrator> # IT Users
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Kabir.IT,CN=Users,DC=task,DC=local" -TargetPath "OU=IT,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Yash.IT,CN=Users,DC=task,DC=local" -TargetPath "OU=IT,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator> Move-ADObject -Identity "CN=Vikram.IT,CN=Users,DC=task,DC=local" -TargetPath "OU=IT,OU=Departments,DC=task,DC=local"
PS C:\Users\Administrator>
```

(Commands to move user in their respective department)



(Users of finance)

Active Directory Users and Computers

FileActionViewHelp

Active Directory Users and Comp

> Saved Queries

task.local

> Builtin

> Computers

> Domain Controllers

> ForeignSecurityPrincipals

> Managed Service Account

Users

Departments

Finance

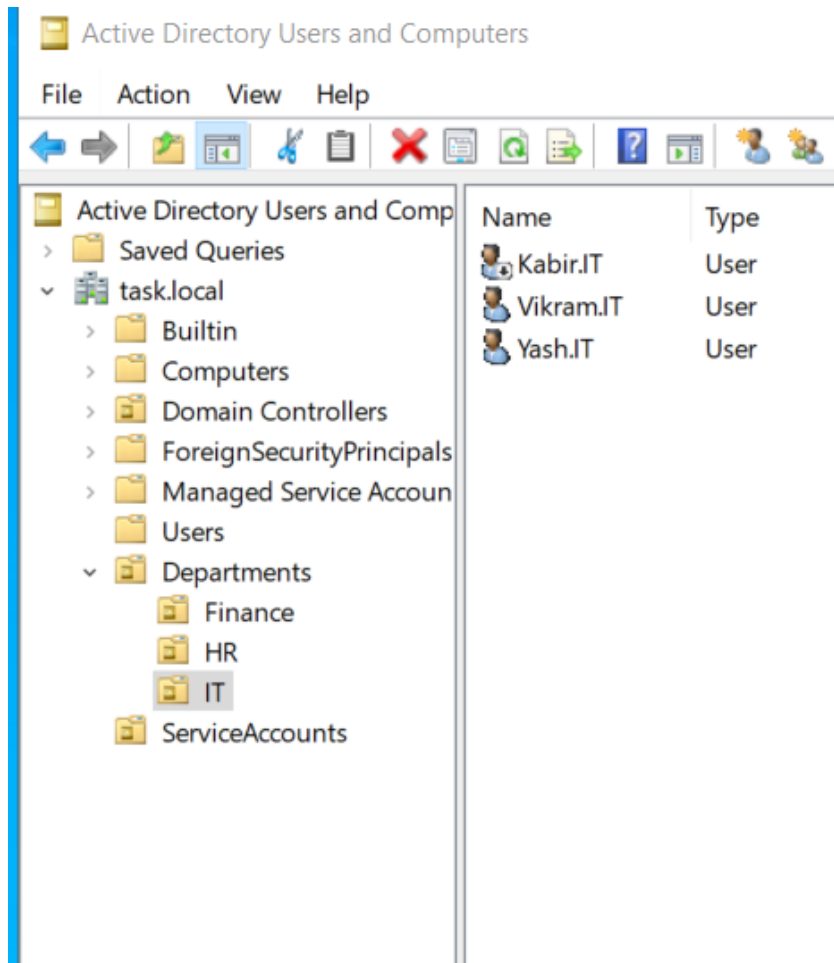
HR

IT

ServiceAccounts

Name	Type	Description
Ananya.HR	User	
Rohan.HR	User	
Sneha.HR	User	

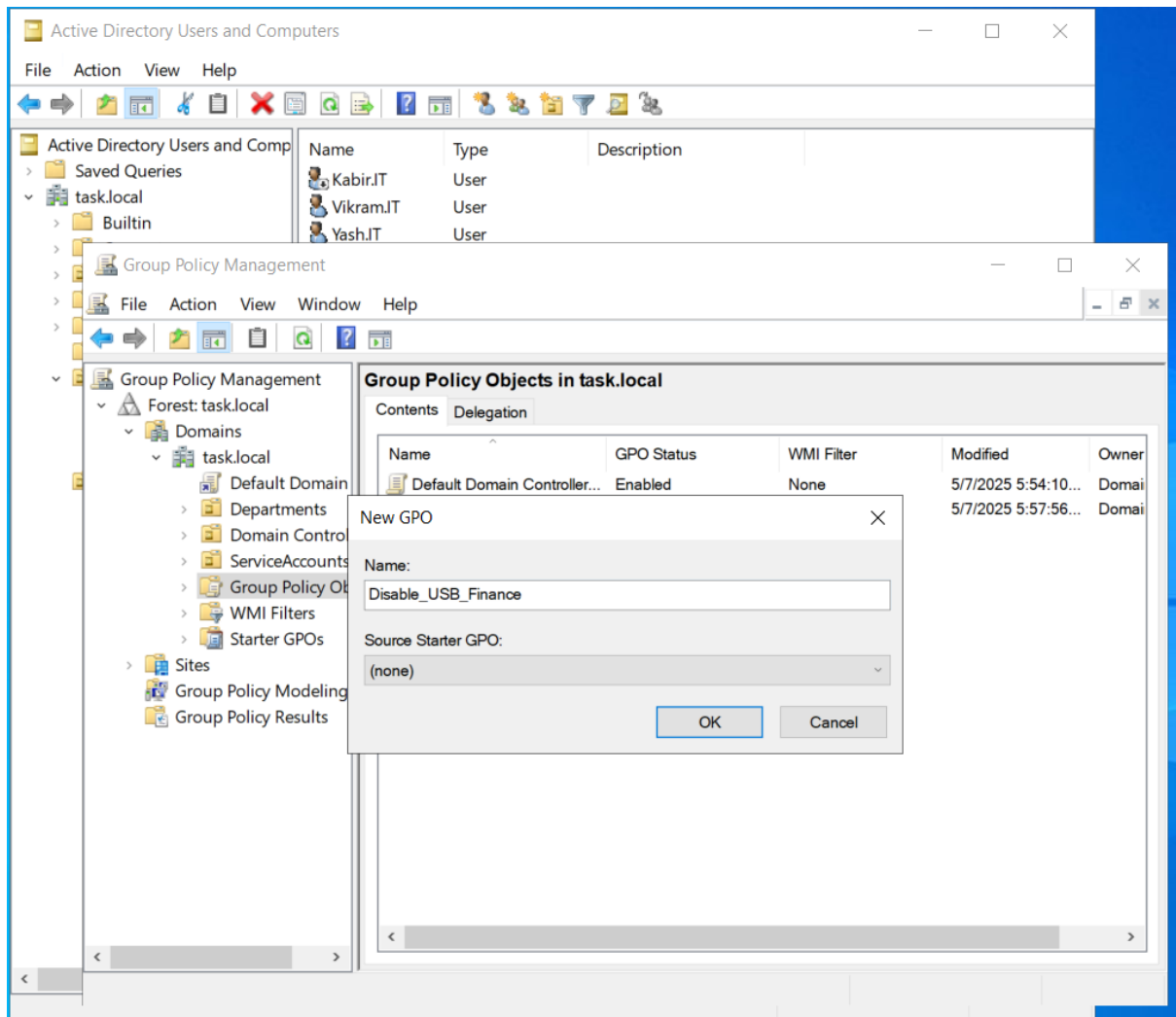
(Users of HR)



(Users of IT)

4.3 Apply GPO to Disable USB Storage for Finance OU

In the Group Policy Management Console, a new GPO named Disable_USB_Finance was created. Under the policy path Computer Configuration → Policies → Administrative Templates → System → Removable Storage Access, the setting "All Removable Storage classes: Deny all access" was configured and set to Enabled. The GPO was then linked to the Finance OU using the GPMC interface.



(Creating a new group policy object)

Group Policy Objects in task.local				
Contents Delegation				
Name	GPO Status	WMI Filter	Modified	Owner
Default Domain Controller...	Enabled	None	5/7/2025 5:54:10...	Domai
Default Domain Policy	Enabled	None	5/7/2025 5:57:56...	Domai
Disable_USB_Finance	Enabled	None	5/10/2025 4:09:2...	Domai

(Successfully created)

All Removable Storage classes: Deny all access

Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on: At least Windows Vista

Options:

Help:

Configure access to all removable storage classes.

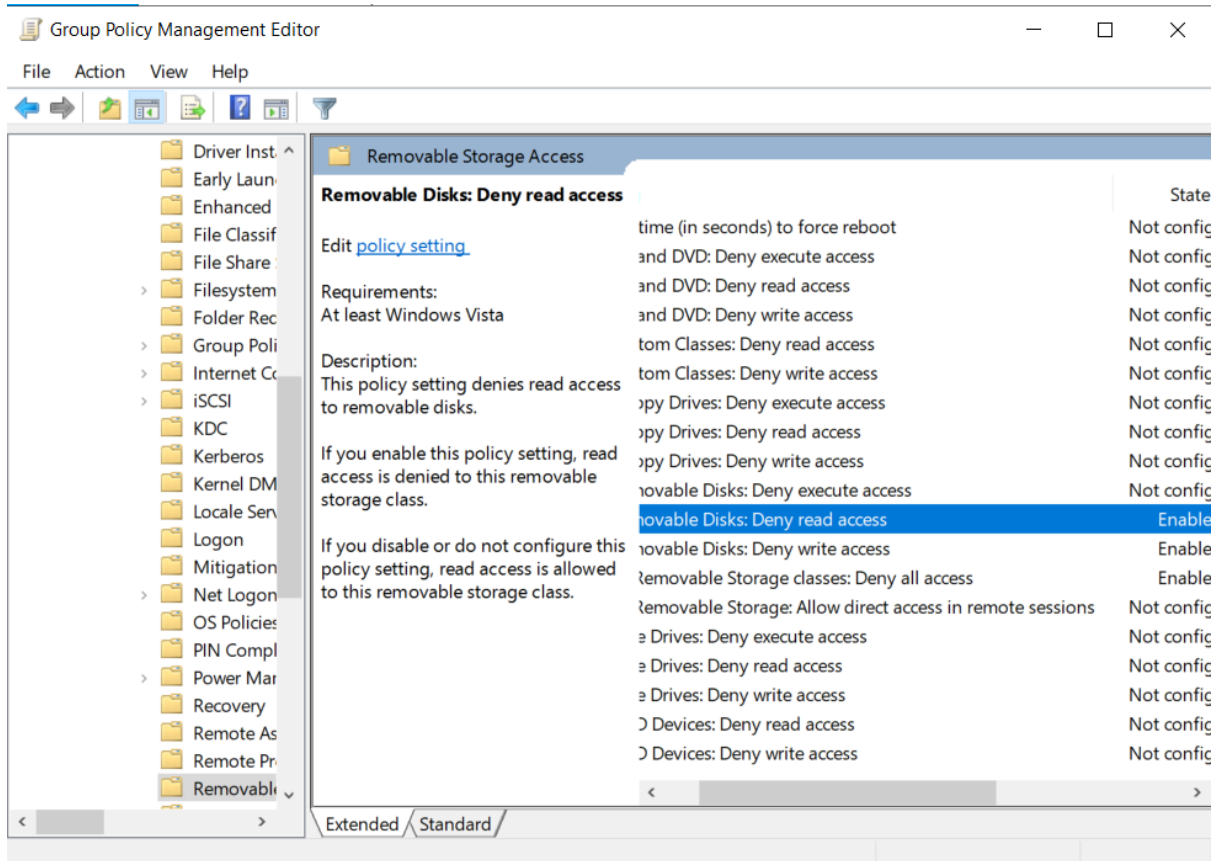
This policy setting takes precedence over any individual removable storage policy settings. To manage individual classes, use the policy settings available for each class.

If you enable this policy setting, no access is allowed to any removable storage class.

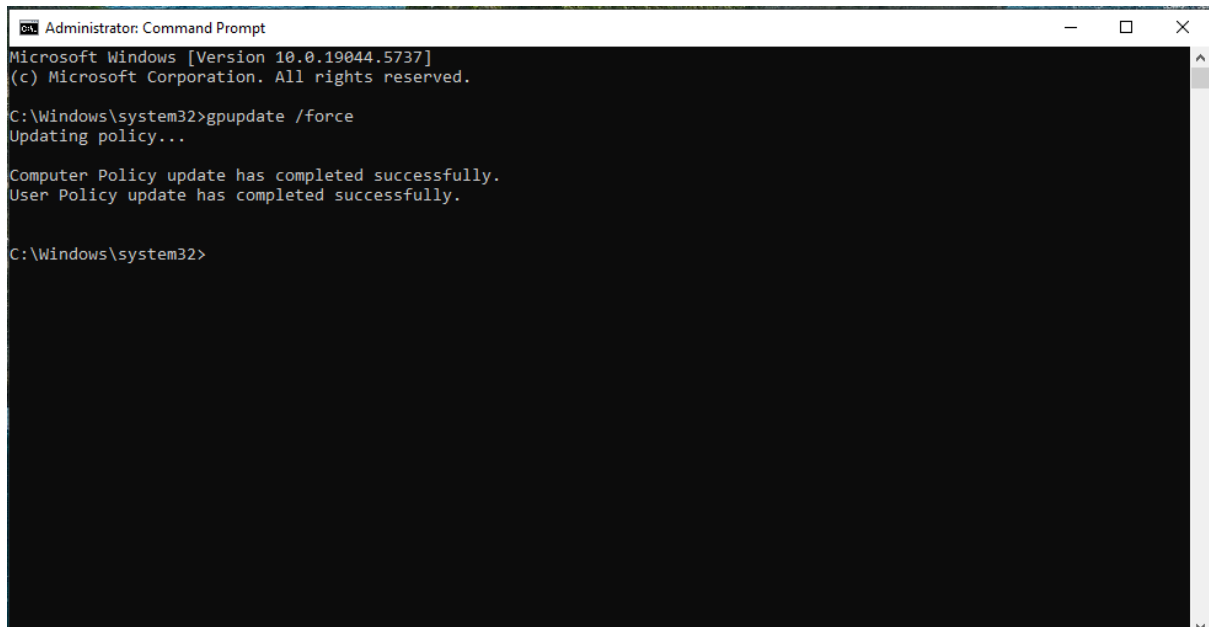
If you disable or do not configure this policy setting, write and read accesses are allowed to all removable storage classes.

OK Cancel Apply

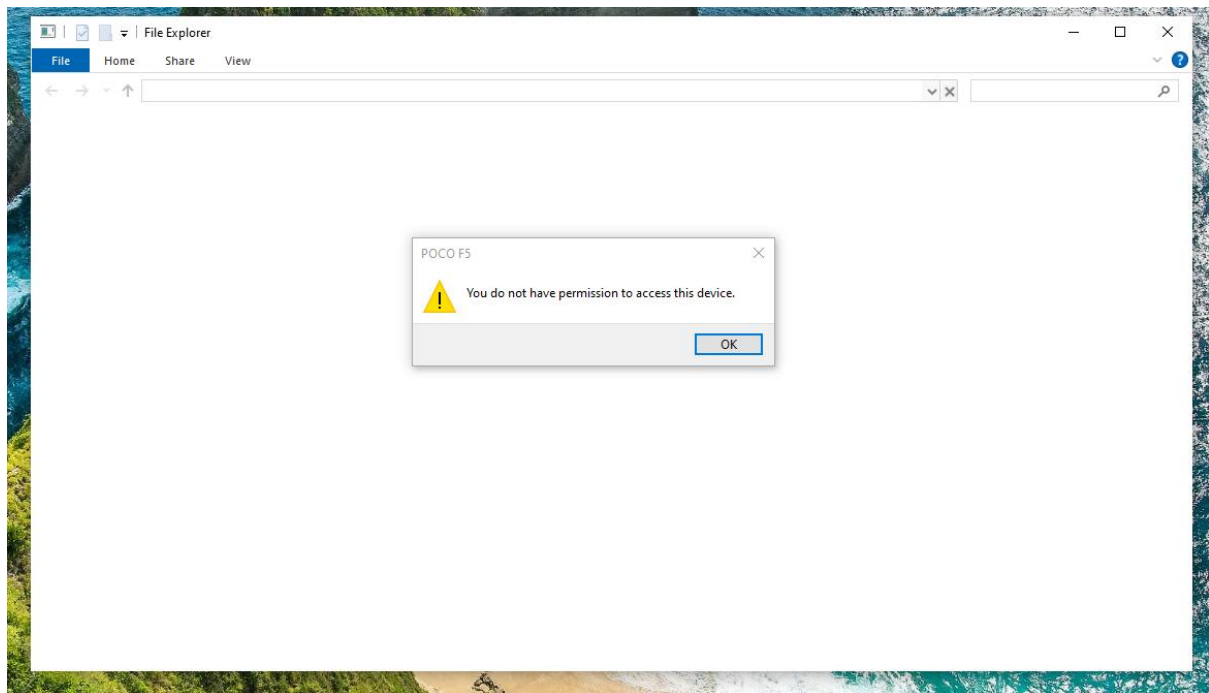
(Denied access to usb)



(Disabled access successfully)



(Policy is updated on client's machine)



(Tried to connect my phone via usb and policy update is successful)

4.4 Verify GPO Application using gpresult

The domain-joined computer PC1 was moved from the default Computers container into the Finance OU. On the client machine, the `gpupdate /force` command was run and the system was rebooted. After logging in, `gpresult /r` was executed in an elevated command prompt. Under Computer Settings, the applied GPO list displayed `Disable_USB_Finance`, confirming that the policy was active for that machine.

```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.19044.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on [ 5/ 10/ 2025 at 5:36:27 PM

RSOP data for TASK\Administrator on PC1 : Logging Mode
-----

OS Configuration:           Member Workstation
OS Version:                 10.0.19044
Site Name:                  Default-First-Site-Name
Roaming Profile:            N/A
Local Profile:              C:\Users\administrator
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=PC1,OU=Finance,OU=Departments,DC=task,DC=local
Last time Group Policy was applied: 5/10/2025 at 5:33:36 PM
Group Policy was applied from:    server1.task.local
Group Policy slow link threshold: 500 kbps
Domain Name:                     TASK
Domain Type:                     Windows 2008 or later

Applied Group Policy Objects
-----
    Disable_USB_Finance
    Default Domain Policy

The following GPOs were not applied because they were filtered out
-----
    Local Group Policy
        Filtering:  Not Applied (Empty)

The computer is a part of the following security groups
-----
    BUILTIN\Administrators
    Everyone
    BUILTIN\Users
    NT AUTHORITY\NETWORK
    NT AUTHORITY\Authenticated Users
    This Organization
    PC1$
    Domain Computers
    Authentication authority asserted identity
    System Mandatory Level

```

(This shows now user device have Disable_USB_Finance GPO)

5. Recommendations

- Create and follow a standard OU structure based on real organizational hierarchy.

- Ensure that both users and computers are placed in correct OUs to apply scoped policies.
- Use group policies with specific targeting to control access per department.
- Confirm each GPO by checking gpresult or HTML report output from affected endpoints.
- Reboot machines after major GPO changes to ensure settings apply correctly.

6. Conclusion

Organizational Units were created for Finance, HR, IT, and service accounts to introduce structure in Active Directory. Users were placed into their appropriate departmental OUs using PowerShell. A USB restriction policy was created and correctly linked to the Finance OU. A domain-joined client machine was moved into the same OU and confirmed to receive the GPO using gpresult. These steps ensured that policies can now be scoped and managed effectively based on organizational structure.